

Global Medical Device Supply Chain Analytics

Root Cause Analysis

Business Problem

Warehouse inventory records do not match the physical inventory count for a critical medical device component.

Problem Statement

During the monthly inventory audit, the ERP system reported 1,250 units, while the warehouse physically contained 1,180 units.

This resulted in:

- Inventory variance
- Procurement planning inaccuracies
- Increased stock-out risk

5 Whys Analysis

Why 1

Why did the ERP inventory differ from the physical inventory?
Because several received items were not recorded correctly.

Why 2

Why were the items not recorded correctly?
Because warehouse staff delayed barcode scanning during peak receiving hours.

Why 3

Why was barcode scanning delayed?
Because incoming deliveries exceeded the available receiving capacity.

Why 4

Why did receiving capacity become insufficient?
Because labor scheduling did not account for periods of high inbound volume.

Why 5

Why was labor planning inaccurate?
Because historical shipment trends were not used when planning warehouse staffing.

Root Cause

Receiving capacity planning was not aligned with forecasted inbound shipment volumes, leading to delayed inventory transactions.

Recommended Improvements

- Schedule warehouse labor using historical inbound trends.
- Implement real-time barcode scanning at receiving docks.
- Introduce daily inventory reconciliation for high-value materials.
- Create automated alerts for delayed Goods Receipt postings.
- Review receiving capacity weekly.

Expected Benefits

- Improved inventory accuracy.
- Reduced stock variance.
- More reliable demand planning.
- Higher confidence in ERP inventory data.
- Better decision-making through accurate Power BI reporting.

Disclaimer: *This project was developed as a personal portfolio project using a simulated supply chain dataset. The company context, business scenarios, and analysis are intended to demonstrate Business Intelligence, data modeling, and dashboard development skills and do not represent actual operational data.*